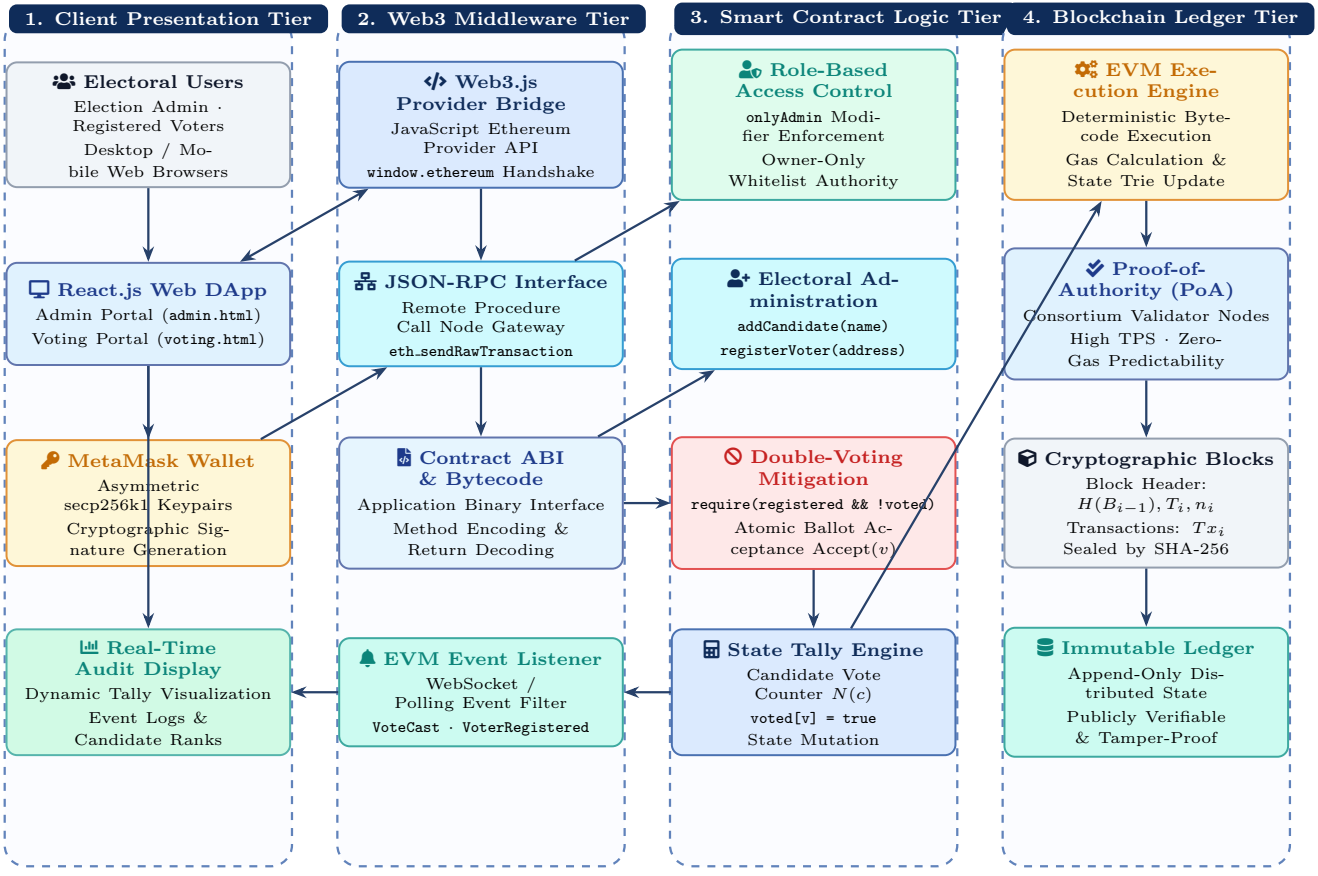

1. Client Presentation Tier
2. Web3 Middleware Tier
3. Smart Contract Logic Tier
4. Blockchain Ledger Tier
Electoral Users
Election Admin · Registered Voters
Desktop / Mobile Web Browsers
Web3.js Provider Bridge
JavaScript Ethereum Provider API
window.ethereum Handshake
Role-Based Access Control
onlyAdmin Modifier Enforcement
Owner-Only Whitelist Authority
EVM Execution Engine
Deterministic Bytecode Execution
Gas Calculation & State Trie Update
React.js Web DApp
Admin Portal (admin.html)
Voting Portal (voting.html)
JSON-RPC Interface
Remote Procedure Call Node Gateway
eth_sendRawTransaction
Electoral Administration
addCandidate(name)
registerVoter(address)
Proof-of-Authority (PoA)
Consortium Validator Nodes
High TPS · Zero-Gas Predictability
MetaMask Wallet
Asymmetric secp256k1 Keypairs
Cryptographic Signature Generation
Contract ABI & Bytecode
Application Binary Interface
Method Encoding & Return Decoding
Double-Voting Mitigation
require(registered && !voted)
Atomic Ballot Acceptance Accept(v)
Cryptographic Blocks
Block Header: $H(B_{i-1}), T_i, n_i$
Transactions: Tx_i
Sealed by SHA-256
Real-Time Audit Display
Dynamic Tally Visualization
Event Logs & Candidate Ranks
EVM Event Listener
WebSocket / Polling Event Filter
VoteCast · VoterRegistered
State Tally Engine
Candidate Vote Counter $N(c)$
voted[v] = true
State Mutation
Immutable Ledger
Append-Only Distributed State
Publicly Verifiable & Tamper-Proof